

Frank Mohn Services AS

Service Bulletin

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Planned Maintenance System - Updated Recommendations



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Dear Customer!

RECOMMENDATIONS TO REDUCE YOUR MAINTENANCE COSTS

With the implementation of the ISM-code, a planned maintenance program is required. In this respect we would like to draw your attention to the part of the program concerning the FRAMO cargo pumping plant. Based on experience, new technology and feedback from our customers, the maintenance recommendations have changed over time.

We recommend you to update your program according to this information.



Hydraulic Oil Filters - Change Intervals

In the original instruction manuals you may find that the filters have been recommended changed once a year, even without any filter alarm. Today the filters are to be changed based on condition only, either from a filter alarm due to high differential filter pressure or from the analysis result of oil samples.

Should no filter alarm occur and your hydraulic oil is in good condition (cleanliness level better than ISO code 16/12) the filters should not be changed.

We recommend that you draw a sample for particle contamination analysis four times a year. In addition, a full analysis should be carried out once a year. FRAMO can offer assistance during oil sampling, monitoring and trend analysis as well as providing advice of preventive maintenance.



Reduced Water Content in Hydraulic Oil - Increased Lifetime

Previously the maximum allowed water content in the hydraulic oil was 0,1%. Experience has shown that a low water content in the hydraulic oil improves lifetime of hydraulic components as well as of the hydraulic oil itself. We have found that the maximum water content should be below 0,05% or 500ppm.

The water content should normally be well below 300ppm.

Any increase in water content must be investigated, the cause corrected and the water removed. To remove water we may offer rental of purifier and / or special water filter. Note that this is only a temporary solution - the cause of the water ingress must be located and repaired!

Testing of Cargo Pumps

In the past we have recommended you to test run your cargo pumps against closed discharge valve, once or twice a year. Such a test requires calibrated pressure gauges and there are several possibilities for error as well as the amount of work involved. Provided no problems are experienced during discharge, parallel pumping or stripping, there is no need for periodic testing of the cargo pumps.

The cargo pump test is useful when trouble shooting a specific pump.

Testing of cargo pumps is recommended in view of a planned docking.



Pre-docking tests are necessary to set up a proper repair schedule to determine which cargo pumps will need an overhaul.

Cargo pump testing is also carried out during on board training courses.

This is to inform the crew of how to do the test and learn to evaluate the results.

In view of above we kindly ask you to change this activity from periodic to condition based in your planned maintenance program.

Framo Tool Boxes - Use Special Tools and Reduce Your Service Costs

Experience has shown that the originally supplied special tools contained in the FRAMO tool boxes are often lost or damaged over the years. Doing maintenance on the cargo pumps without these tools may cause damage to the pump as well as to the new parts, increasing your total costs. It is better to renew the contents of the tool box than paying for damage during repair or even malfunction during the following discharge operation.

Please include an annual check of the FRAMO tool box in your maintenance program.

Spline Lubrication

Experience indicates that inspection and lubrication of the spline connection between the hydraulic pumps and the PTO output shafts is neglected, causing abnormal wear on the spline. This is also the case on electric motors with spline connection.

It is very important to have a scheduled routine for control of this lubrication. We recommend that the oil on the spline is changed every 800 running hours, or every six months, whichever comes first.



Level control



If you need any further advices or assistance, please do not hesitate to contact us!



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